Digital Communication IC Design

Homework 2

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Development Environment

Language	MATLAB R2025b
AI Tool	Claude Opus 5.5

Due Date: 2026/10/21

Contents

1 Student-Specific Parameters **2**

2 Q1 — Zadoff-Chu Sequence **3**

2.1 (a) Root-u Zadoff-Chu Sequence 3

2.2 (b) Periodic and Linear Autocorrelation 5

2.3 (c) Periodic Cross-Correlation 7

2.4 (d) Root and Delay Detection 8

3 Q2 — Barker Code Periodic Autocorrelation **10**

4 Q3 — m-Sequences and Gold Code **12**

4.1 (a) Maximal-Length Sequences 12

4.2 (b) Periodic Autocorrelation 14

4.3 (c) Gold Codeword and Cross-Correlation 15

4.4 (d) Is the Answer in (c) Correct? 17

5 AI Tool Disclosure **19**

Student-Specific Parameters

My student ID is M11407439, so $m = 39$. The table lists the parameters I get from it.

Table 1.1: Parameters Derived from $m = 39$

Quantity	**Definition**	**Value**
ZC root u	$1 + (m \bmod 52)$	$1 + 39 = \mathbf{40}$
ZC root v	$1 + ((u + 16) \bmod 52)$	$1 + (56 \bmod 52) = \mathbf{5}$
Transmitted root u_0	$3 + ((7m + 2) \bmod 48)$	$3 + (275 \bmod 48) = \mathbf{38}$
Transmitted delay τ_0	$(5m + 3) \bmod 48$	$198 \bmod 48 = \mathbf{6}$
Candidate roots Ω	$\{u_0 - 2, u_0, u_0 + 2\}$	$\{\mathbf{36}, \mathbf{38}, \mathbf{40}\}$
LFSR state r_1	$1 + (m \bmod 20)$	$20 = (010100)_2$
LFSR state r_2	$11 + (m \bmod 20)$	$30 = (011110)_2$

The code is in `01_src`. `q1.m`, `q2.m` and `q3.m` handle the three questions, and `new_fig.m`, `save_fig.m`, `tl_labels.m` and `hw_palette.m` only control how the figures look. The numbers quoted in this report are printed by the scripts when they run.

Q1 — Zadoff-Chu Sequence

2.1 (a) Root-u Zadoff-Chu Sequence

Use the prime sequence length $N = 53$. Let $u = 1 + (m \bmod 52)$, where m is the last two digits of the student ID. For odd N, define the root-u Zadoff-Chu sequence as

$$x_u[n] = e^{-j\pi un(n+1)/N}, \quad n = 0, \ldots, N-1.$$

Generate $x_u[n]$. Plot its real part, imaginary part, magnitude, and unwrapped phase. (10%)

With $u = 40$ and $N = 53$, I compute $x_{40}[n]$ straight from the definition. $\gcd(40, 53) = 1$, so 40 is a valid root. The first samples are $x_{40}[0] = 1$, $x_{40}[1] = 0.0296 + 0.9996j$ and $x_{40}[2] = -0.0888 - 0.9960j$.

The magnitude panel is flat: $|x_u[n]| = 1$ for every n, up to a rounding error of 1.1×10^{-16}. All the information sits in the phase. Constant amplitude plus the zero autocorrelation shown in (b) is what people call a CAZAC sequence, and it is the reason LTE uses ZC sequences for PSS and PRACH.

The real and imaginary parts are also mirror images about $n = 26$. This is easy to prove. Since $(N-1-n)(N-n) = n(n+1) + N(N-1-2n)$, we have $x_u[N-1-n] = x_u[n]\, e^{-j\pi u(N-1-2n)}$, and $N - 1 - 2n$ is even, so $x_u[N-1-n] = x_u[n]$.

The unwrapped phase needs a comment, because it does not look like the parabola $\phi[n] = -\pi un(n+1)/N$ in the definition. That parabola drops to -6534.5 rad at $n = 52$, and its step from one sample to the next is $-2\pi u(n+1)/N$, which is already larger than π at $n = 0$. MATLAB `unwrap` assumes each step is in $(-\pi, \pi]$, so it cannot follow the parabola. It picks the path with the smallest steps instead, which stays between 0 and 13.2 rad and keeps the same mirror symmetry. The two paths differ only by multiples of 2π (the residue after wrapping is at most 9.5×10^{-13}), so they describe exactly the same samples. In other words, the chirp is aliased by the sampling. The same fact explains why root u and root $u - N = -13$ give the same sequence: $e^{-j\pi(u-N)n(n+1)/N} = x_u[n]\, e^{j\pi n(n+1)} = x_u[n]$, because $n(n+1)$ is always even.

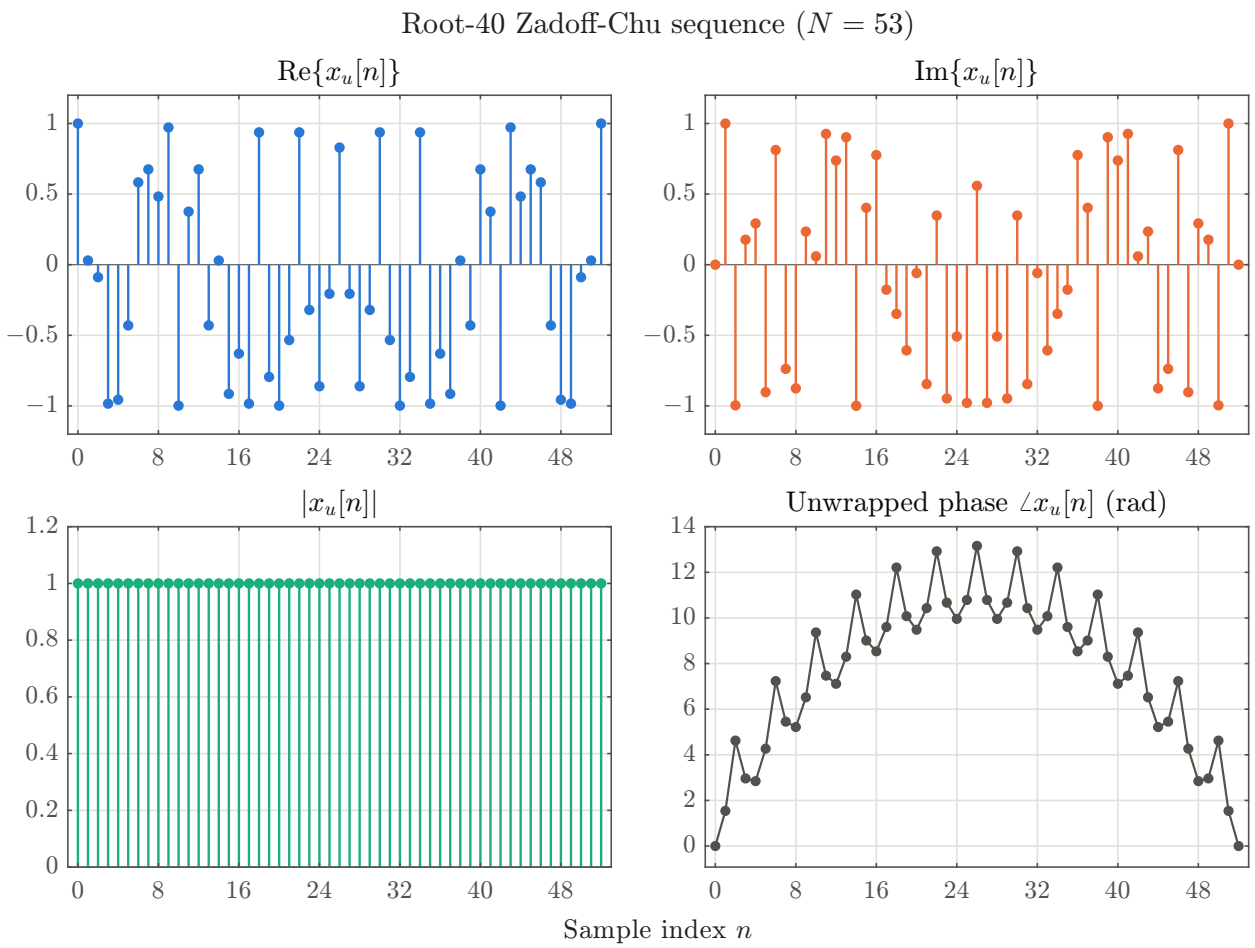


Figure 2.1: Real Part, Imaginary Part, Magnitude and Unwrapped Phase of $x_{40}[n]$

2.2 (b) Periodic and Linear Autocorrelation

Calculate and stem-plot the normalized periodic autocorrelation for $l = 0, 1, \ldots, N-1$. (10%) Also calculate a nonperiodic linear autocorrelation and explain why it does not exhibit the same zero-sidelobe property. (10%)

$$R_u[l] = \frac{1}{N}\sum_{n=0}^{N-1} x_u[n]\, x_u^*[(n-l) \bmod N], \qquad R_u^{\text{lin}}[l] = \frac{1}{N}\sum_{n=l}^{N-1} x_u[n]\, x_u^*[n-l].$$

Periodic autocorrelation. I get $R_u[0] = 1$ and $\max_{l\neq 0} |R_u[l]| = 7.9 \times 10^{-14}$. The small residue comes from rounding the large phase arguments, so all 52 sidelobes are zero.

To see why, multiply a ZC sample by a shifted copy. The n^2 terms cancel and the phase of the product is *linear* in n:

$$x_u[n]\, x_u^*[n-l] = e^{-j\pi u[\, n(n+1)-(n-l)(n-l+1)\,]/N} = \underbrace{e^{-j\pi u(l-l^2)/N}}_{c_l}\; e^{-j2\pi uln/N}. \tag{2.1}$$

This still holds with the cyclic index $(n-l) \bmod N$. Replacing $n-l$ by $n-l+N$ changes the phase by $\pi u(2n-2l+N+1)$, and with N odd that is a multiple of 2π. So

$$R_u[l] = \frac{c_l}{N}\sum_{n=0}^{N-1} e^{-j2\pi uln/N} = c_l\, \delta[l], \tag{2.2}$$

since for $0 < l < N$ the product ul is not a multiple of the prime N, and a complex exponential summed over whole periods gives zero.

Linear autocorrelation. The summand is the same, but the sum now runs over $n = l, \ldots, N-1$ only, which is $N-l$ terms instead of a full period. The geometric series does not close, and

$$|R_u^{\text{lin}}[l]| = \frac{1}{N}\left|\frac{\sin\left(\pi ul(N-l)/N\right)}{\sin(\pi ul/N)}\right|, \quad l \neq 0. \tag{2.3}$$

which is generally not zero. The simulated values agree with this formula to 3.9×10^{-14} (black circles in the figure). The largest sidelobe is $|R_u^{\text{lin}}[8]| = 0.1296$, or -17.7 dB, and the average sidelobe is 0.0278.

So the zero sidelobes depend on the wrap-around. In the periodic sum, the samples that slide off the end of the shifted copy reappear at the start, and the N terms form a full set of N-th roots of unity. The linear sum loses l of those terms, and what is left does not cancel. LTE handles this with a cyclic prefix in front of the ZC sequence, which makes the receiver's linear correlation behave like a periodic one.

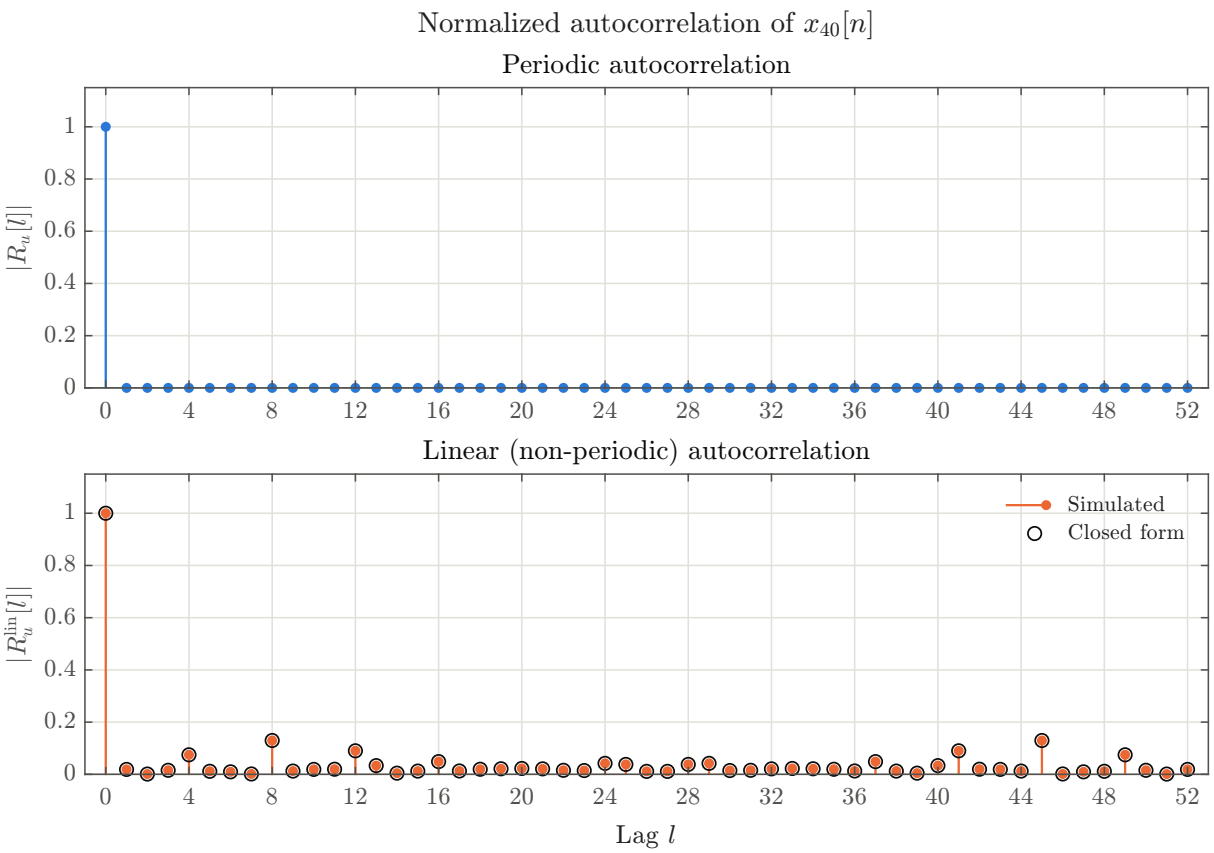


Figure 2.2: Periodic (top) and Linear (bottom) Autocorrelation of $x_{40}[n]$

2.3 (c) Periodic Cross-Correlation

Choose $v = 1 + ((u + 16) \bmod 52)$. Calculate the periodic cross-correlation $C_{u,v}[l]$ for $l = 0, 1, \ldots, N-1$. (10%)

$$C_{u,v}[l] = \frac{1}{N} \sum_{n=0}^{N-1} x_u[n]\, x_v^*[(n-l) \bmod N]$$

With $u = 40$ and $v = 5$, the magnitude of the cross-correlation does not depend on the lag:

$$|C_{40,5}[l]| = 0.137361 = \frac{1}{\sqrt{53}} \quad \text{for all } l = 0, \ldots, 52. \tag{2.4}$$

The minimum and maximum agree to six decimals. Only the phase changes with l (bottom panel).

This time the n^2 terms do not cancel. The phase of $x_u[n]x_v^*[n-l]$ is $-\pi(u-v)n^2/N$ plus a term linear in n, which makes $C_{u,v}[l]$ a quadratic Gauss sum. When N is prime and $\gcd(u-v, N) = \gcd(35, 53) = 1$, such a sum always has magnitude $\sqrt{N}$, so $|C_{u,v}[l]| = \sqrt{N}/N = 1/\sqrt{N}$. A flat level cannot be any lower than this, because Parseval requires $\sum_l |C_{u,v}[l]|^2 = 1$ for unit-energy sequences (here $53 \times 1/53 = 1$). The practical meaning is that a wrong root never looks like a weak copy of the right one. Its correlation is spread evenly over all lags, with no peak anywhere.

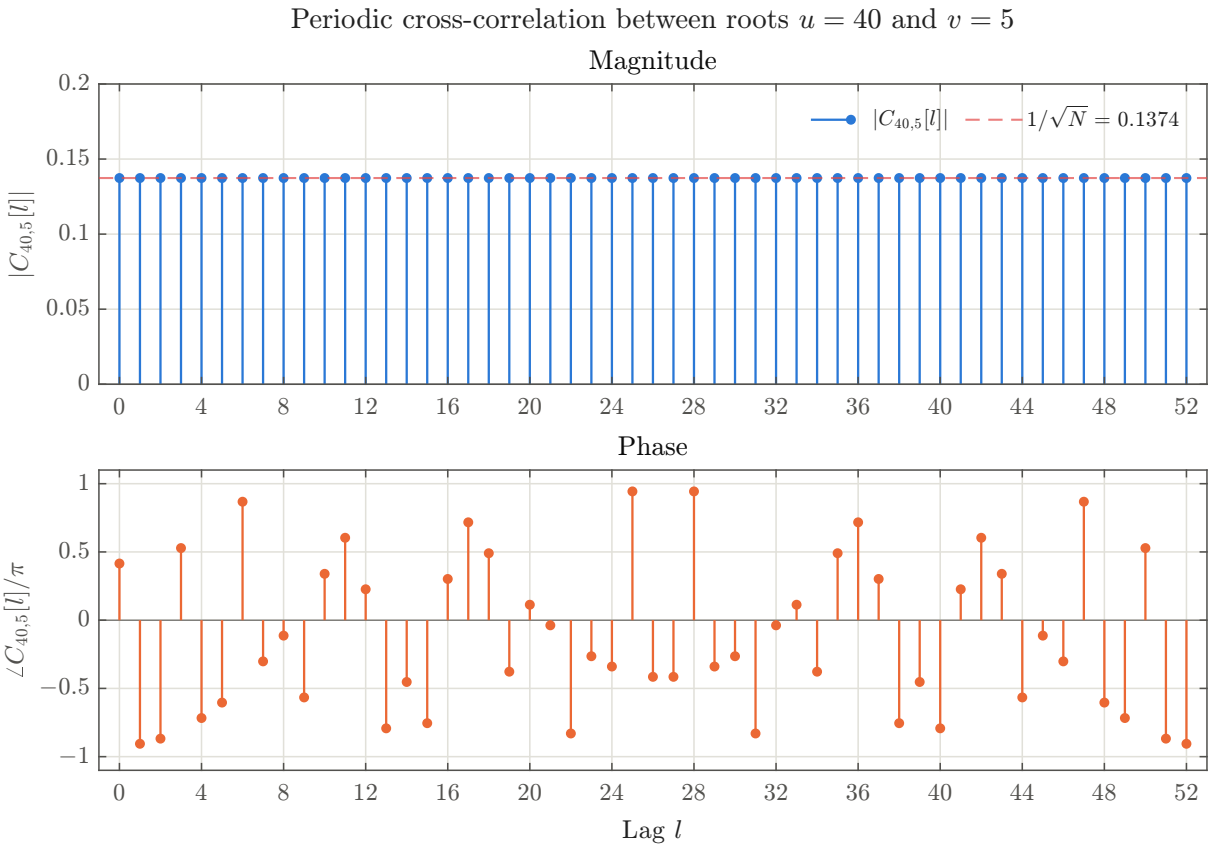


Figure 2.3: Magnitude (top) and Phase (bottom) of the Periodic Cross-Correlation $C_{40,5}[l]$

2.4 (d) Root and Delay Detection

Construct a student-specific synchronization test with $u_0 = 3 + ((7m + 2) \bmod 48)$, $\tau_0 = (5m + 3) \bmod 48$, complex gain $\alpha = \exp(j\pi/5)$. The received signal is defined as $r[n] = \alpha x_{u_0}[(n - \tau_0) \bmod N]$, $n = 0, 1, \ldots, N - 1$. If the set $\Omega = \{u_0 - 2, u_0, u_0 + 2\}$, search three possible roots in set Ω and 53 cyclic delays using the metric below. For each root in the set Ω, draw $D(u, \tau)$ versus τ. (15%) Report the detected root and delay, and whether they match u_0 and τ_0. Please add some explanations. (5%)

$$D(u, \tau) = \left| \frac{1}{N} \sum_{n=0}^{N-1} r[n]\, x_u^*[(n - \tau) \bmod N] \right| \quad \text{for } u \in \Omega \text{ and } \tau = 0, 1, \ldots, 52$$

With $u_0 = 38$ and $\tau_0 = 6$, the received signal is $r[n] = e^{j\pi/5} x_{38}[(n - 6) \bmod 53]$ and the search set is $\Omega = \{36, 38, 40\}$. I compute all $3 \times 53 = 159$ values of $D(u, \tau)$ and pick the largest.

Table 2.1: Detection Metric Summary

u	$\max_\tau D(u, \tau)$	$\arg\max_\tau$	$D(u, \tau)$ at other lags
36	0.1374	– (flat)	0.1374 at all 53 lags
38	**1.0000**	$\boldsymbol{\tau = 6}$	0 (all 52 other lags)
40	0.1374	– (flat)	0.1374 at all 53 lags

Result. The largest value is $D(38, 6) = 1$, so the detected pair is $(\hat{u}, \hat{\tau}) = (38, 6)$. It **matches** $(u_0, \tau_0) = (38, 6)$. The correlation at the peak has phase 0.2000π, which means the same search also gives the channel phase $\angle\alpha = \pi/5$.

Explanation. The three curves are what (b) and (c) predict. For the correct root $u = 38$, $D(u_0, \tau)$ is $|\alpha|$ times the periodic autocorrelation of x_{38} at lag $\tau - \tau_0$. From (b), that is $|\alpha| = 1$ at $\tau = \tau_0$ and zero at every other delay, so the peak is a single stem and the delay is unambiguous.

For the wrong roots $u = 36$ and 40, $D(u, \tau)$ is a cross-correlation between two different roots. The difference $u - u_0 = \mp 2$ is coprime to 53, so by (c) the value is $1/\sqrt{N} = 0.1374$ at every delay. These two curves are completely flat. Their "argmax" would just be whichever lag happens to win by rounding error, so the table lists them as flat instead.

The highest wrong value is 0.1374, which puts the true peak 7.28 times (17.2 dB) above everything else. The complex gain has no effect on this: $|\alpha| = 1$ scales all values by the same amount, and its phase is removed by the absolute value. In an LTE receiver doing PSS or PRACH detection, the root would tell which cell or preamble was sent and the delay would give the timing. Noise would raise the flat floor somewhat, but with a 17 dB margin the decision should hold well into low SNR.

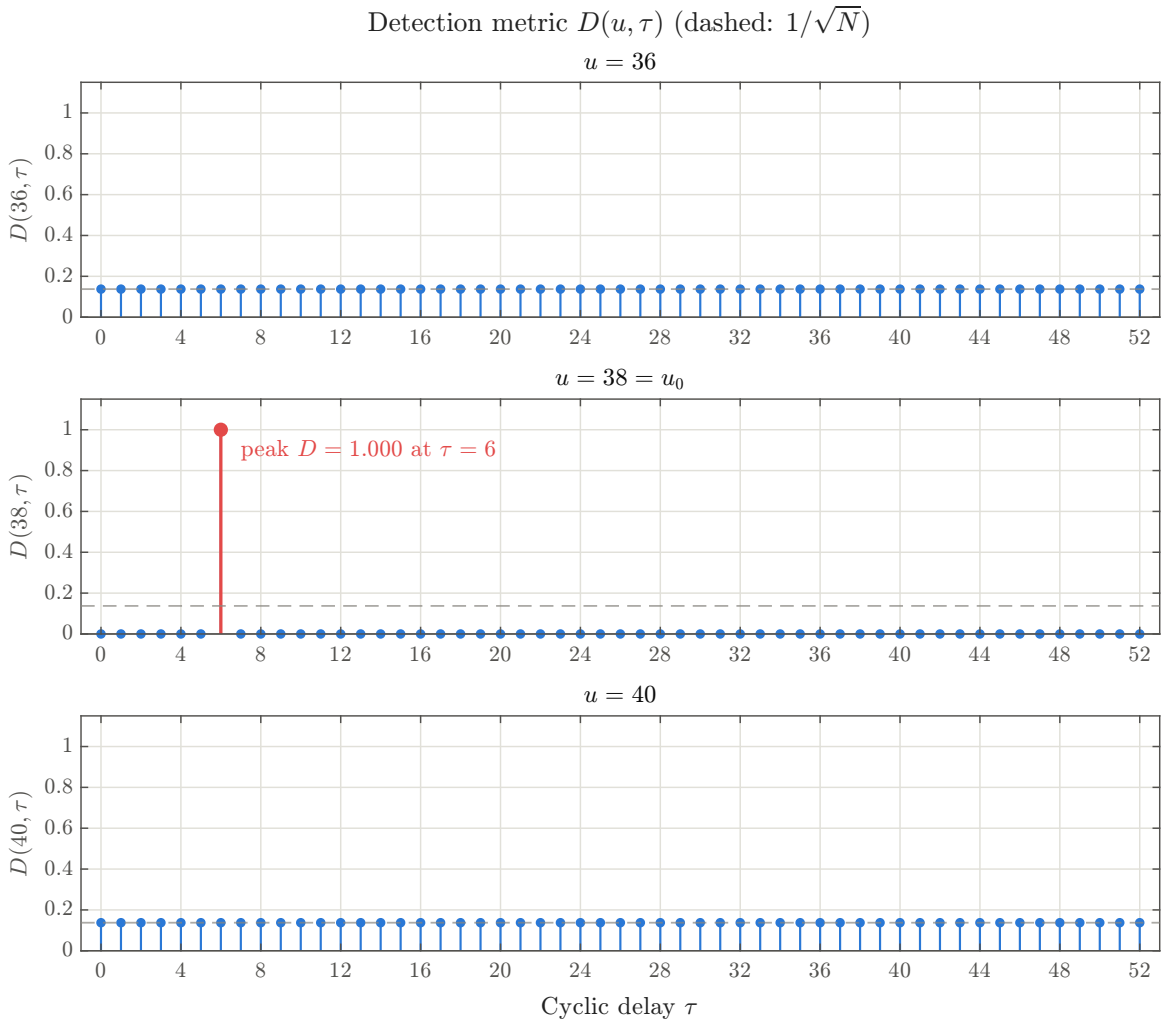


Figure 2.4: Detection Metric $D(u, \tau)$ for $u = 36$, 38 and 40 (red: detected peak)

Q2 — Barker Code Periodic Autocorrelation

Given the Barker code list in Table 1, please check the periodic autocorrelation function defined as

$$\Phi_s(k) = \frac{1}{N} \sum_{n=0}^{N-1} S(n)\, S^*((n-k) \bmod N)$$

for $N = 7$ and 13. (15%)

I use the two codes as they appear in Table 1 of the spec:

$$N = 7: \quad [-1\ -1\ -1\ +1\ +1\ -1\ +1],$$
$$N = 13: \quad [-1\ -1\ -1\ -1\ -1\ +1\ +1\ -1\ -1\ +1\ -1\ +1\ -1].$$

Table 3.1: $N\Phi_s(k)$ for the Length-7 Barker Code (aperiodic values for reference)

k	0	1	2	3	4	5	6
$N\Phi_s(k)$	7	−1	−1	−1	−1	−1	−1
Aperiodic $\times N$	7	0	−1	0	−1	0	−1

Table 3.2: $N\Phi_s(k)$ for the Length-13 Barker Code (aperiodic values for reference)

k	0	1	2	3	4	5	6	7	8	9	10	11	12
$N\Phi_s(k)$	13	1	1	1	1	1	1	1	1	1	1	1	1
Aperiodic $\times N$	13	0	1	0	1	0	1	0	1	0	1	0	1

Result. Both periodic autocorrelations take only two values, the peak at $k = 0$ and one constant sidelobe:

$$\Phi_s(k) = \begin{cases} 1, & k = 0 \\ -1/7, & k \neq 0 \end{cases} \quad (N = 7), \qquad \Phi_s(k) = \begin{cases} 1, & k = 0 \\ +1/13, & k \neq 0 \end{cases} \quad (N = 13). \tag{3.1}$$

Why. Barker codes are defined through the *aperiodic* autocorrelation $A(k) = \sum_{n=k}^{N-1} S(n)S(n-k)$, with every sidelobe satisfying $|A(k)| \leq 1$. For odd N the sidelobes are 0 at odd k and ±1 at even k, as the second row of each table shows. The periodic sum is the aperiodic one plus the part that wraps around:

$$N\Phi_s(k) = A(k) + A(N-k). \tag{3.2}$$

With N odd, k and $N-k$ always have opposite parity. One of the two terms is therefore 0 and the other is ±1, which is why every periodic sidelobe comes out the same. In the length-7 code the even-k sidelobes are all −1, giving −1/7. In the length-13 code they are all +1, giving +1/13.

Sum check. Adding up all shifts gives $\sum_k N\Phi_s(k) = \left(\sum_n S(n)\right)^2$. For $N = 7$, $\sum S = -1$ and $7 + 6 \times (-1) = 1 = (-1)^2$. For $N = 13$, $\sum S = -5$ and $13 + 12 \times 1 = 25 = (-5)^2$. Put another way, a two-valued result forces the sidelobe to be $\big((\sum S)^2 - N\big)/\big(N(N-1)\big)$, and the script gets this value for both codes.

The periodic sidelobes are only $1/N$ in magnitude, so a Barker code still has one sharp peak when it repeats. That is what makes it useful as a synchronization word; IEEE 802.11b uses the 11-chip version for this.

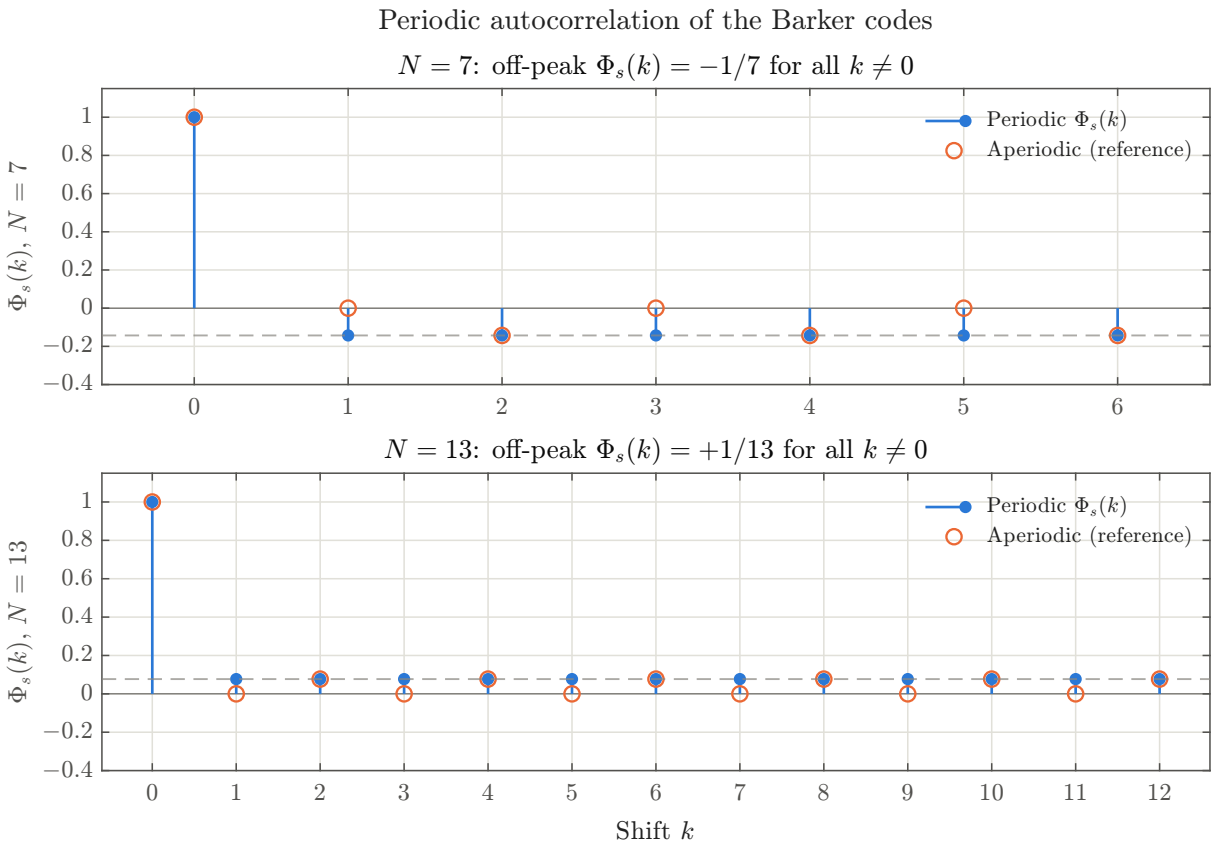


Figure 3.1: Periodic Autocorrelation $\Phi_s(k)$ of the Barker Codes for $N = 7$ (top) and $N = 13$ (bottom); dashed line: off-peak value

Q3 — m-Sequences and Gold Code

The octal pair 103_8 and 147_8 gives

$$103_8 = (1\,000\,011)_2 \;\Rightarrow\; p(x) = x^6 + x + 1, \qquad 147_8 = (1\,100\,111)_2 \;\Rightarrow\; p'(x) = x^6 + x^5 + x^2 + x + 1. \quad (4.1)$$

LFSR connection. I wire the registers as in the spec's figure. Each one has six stages, numbered 1 on the left to 6 on the right. On each clock, the XOR of the tapped stages goes into stage 1, everything shifts one stage to the right, and stage 6 is the output. A stage k is tapped when x^k is in the polynomial, which matches the arrows in the figure:

- top LFSR, $p(x) = x^6 + x + 1$, taps at stages 1 and 6 (two arrows);
- bottom LFSR, $p'(x) = x^6 + x^5 + x^2 + x + 1$, taps at stages 1, 2, 5 and 6 (four arrows).

The generator output is the XOR of the two stage-6 outputs.

Initial states. I write $r_1 = 20$ and $r_2 = 30$ in six-bit binary and put the MSB in stage 1, which gives $r_1 \to$ `010100` and $r_2 \to$ `011110`. Neither is all zeros, so both registers will run.

4.1 (a) Maximal-Length Sequences

Set the initial states r_1 into the top LFSR and set all 0's as the initial state in the bottom LFSR. Generate the Maximal-length sequences $s_1[n]$ of 63 bits, for $n = 0, 1, \ldots, 62$. Print the initial values, the complete sequences. Then, set the initial states r_2 into the bottom LFSR and set all 0's as the initial state in the top LFSR. Generate the Maximal-length sequences $s_2[n]$ of 63 bits, for $n = 0, 1, \ldots, 62$. Print the initial values, the complete sequences. (10%)

A register loaded with all zeros never leaves that state, so in each case the output is simply the loaded register's m-sequence. The script also checks that the empty register outputs nothing but zeros.

Table 4.1: Initial Register Values

	$s_1[n]$	$s_2[n]$
Top LFSR initial state	`010100`	`000000`
Bottom LFSR initial state	`000000`	`011110`

Table 4.2: Complete m-Sequences $s_1[n]$ and $s_2[n]$ (9 bits per row, n increasing left to right)

Sequence	n	Bits
$s_1[n]$	0–8	`001010001`
	9–17	`100001000`
	18–26	`001111110`
	27–35	`101011001`
	36–44	`101110110`
	45–53	`100100111`
	54–62	`000101111`
$s_2[n]$	0–8	`011110000`
	9–17	`011011100`
	18–26	`110001110`
	27–35	`101111110`
	36–44	`110100010`
	45–53	`000101100`
	54–62	`101010010`

Checks. The first six output bits are the initial state read backwards, from stage 6 to stage 1. For example, $s_1[0..5] = $ `001010` is `010100` reversed. Each register goes through all $2^6 - 1 = 63$ nonzero states before it comes back to the start, so the period is exactly 63 and both polynomials are primitive. Each sequence has 32 ones and 31 zeros, as an m-sequence should.

One detail about the convention. Because the output is taken from stage 6, the taps in the figure give the recurrence $s_1[n + 6] = s_1[n + 5] \oplus s_1[n]$. Its characteristic polynomial is $x^6 + x^5 + 1$, the reciprocal of $p(x)$. A reciprocal primitive polynomial is still primitive and produces the same m-sequence run backwards in time. Reversing both sequences turns lag l into lag $-l$ and changes nothing else, so the correlation results below do not depend on this choice.

4.2 (b) Periodic Autocorrelation

Map a binary sequence $s_i[n]$ to $d_i[n] = (-1)^{s_i[n]}$. For both sequences, verify the normalized periodic autocorrelation for all 63 lags. (20%)

$$R_i[l] = \frac{1}{63} \sum_{n=0}^{62} d_i[n]\, d_i[(n+l) \bmod 63]$$

For both sequences the simulation gives exactly

$$R_1[l] = R_2[l] = \begin{cases} 1, & l = 0 \\ -1/63 = -0.0159, & l = 1, \ldots, 62. \end{cases} \tag{4.2}$$

This is the two-valued autocorrelation expected of an m-sequence, and it comes from two properties:

1) Shift-and-add. For $l \neq 0$, $s[n] \oplus s[n+l]$ satisfies the same linear recurrence as $s[n]$, so it is another cyclic shift $s[n+k]$ of the same sequence. In ±1 form this reads $d[n]\, d[n+l] = d[n+k]$. The script confirms it for all 62 nonzero lags of d_1.

2) Balance. One period has 32 ones and 31 zeros, so $\sum_n d[n] = 31 - 32 = -1$.

Putting them together, $63\, R[l] = \sum_n d[n+k] = -1$ for every $l \neq 0$.

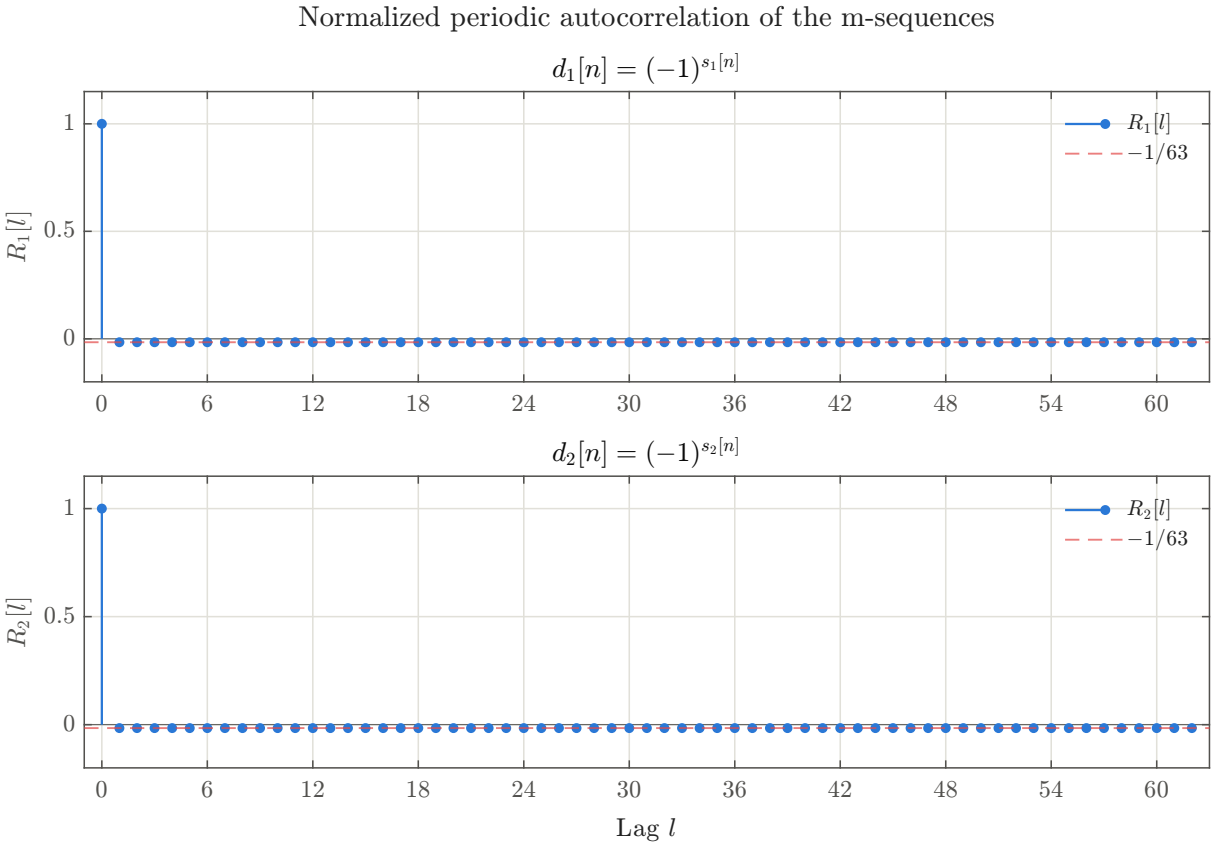


Figure 4.1: Normalized Periodic Autocorrelation of $d_1[n]$ (top) and $d_2[n]$ (bottom)

4.3 (c) Gold Codeword and Cross-Correlation

Now, we will construct another codeword of the Gold code. Put initial state r_1 and r_2 into top LFSR and bottom LFSR, respectively. Generate the output $s_3[n]$ of 63 bits. Now, map $s_3[n]$ to $d_3[n]$. Then, let's check the cross-correlation property for $(u, v) = (1, 3)$ and $(u, v) = (2, 3)$ with $N = 63$. (20%)

$$C_{u,v}[l] = \frac{1}{N} \sum_{n=0}^{N-1} d_u[n]\, d_v[(n - l) \bmod N]$$

Loading `010100` into the top register and `011110` into the bottom one gives the following codeword.

Table 4.3: Gold Codeword $s_3[n]$

Sequence	n	Bits
$s_3[n]$	0–8	`010100001`
	9–17	`111010100`
	18–26	`111110000`
	27–35	`000100111`
	36–44	`011010100`
	45–53	`100001011`
	54–62	`101111101`

The two registers run independently, so $s_3[n] = s_1[n] \oplus s_2[n]$ (the script compares all 63 bits), and in ±1 form $d_3[n] = d_1[n]\, d_2[n]$.

Both cross-correlations take only three values. Table 4.4 counts how many lags give each one.

Table 4.4: Values of 63 $C_{u,v}[l]$ over All 63 Lags

	-17	-1	15	$\max_l \lvert C_{u,v}[l] \rvert$
$C_{1,3}$	6	47	10	$17/63 = 0.2698$
$C_{2,3}$	6	47	10	$17/63 = 0.2698$

So $C_{1,3}[l]$ and $C_{2,3}[l]$ both lie in $\{-17/63,\ -1/63,\ 15/63\}$. The largest magnitude is $17/63 = 0.27$, compared with an autocorrelation peak of 1.

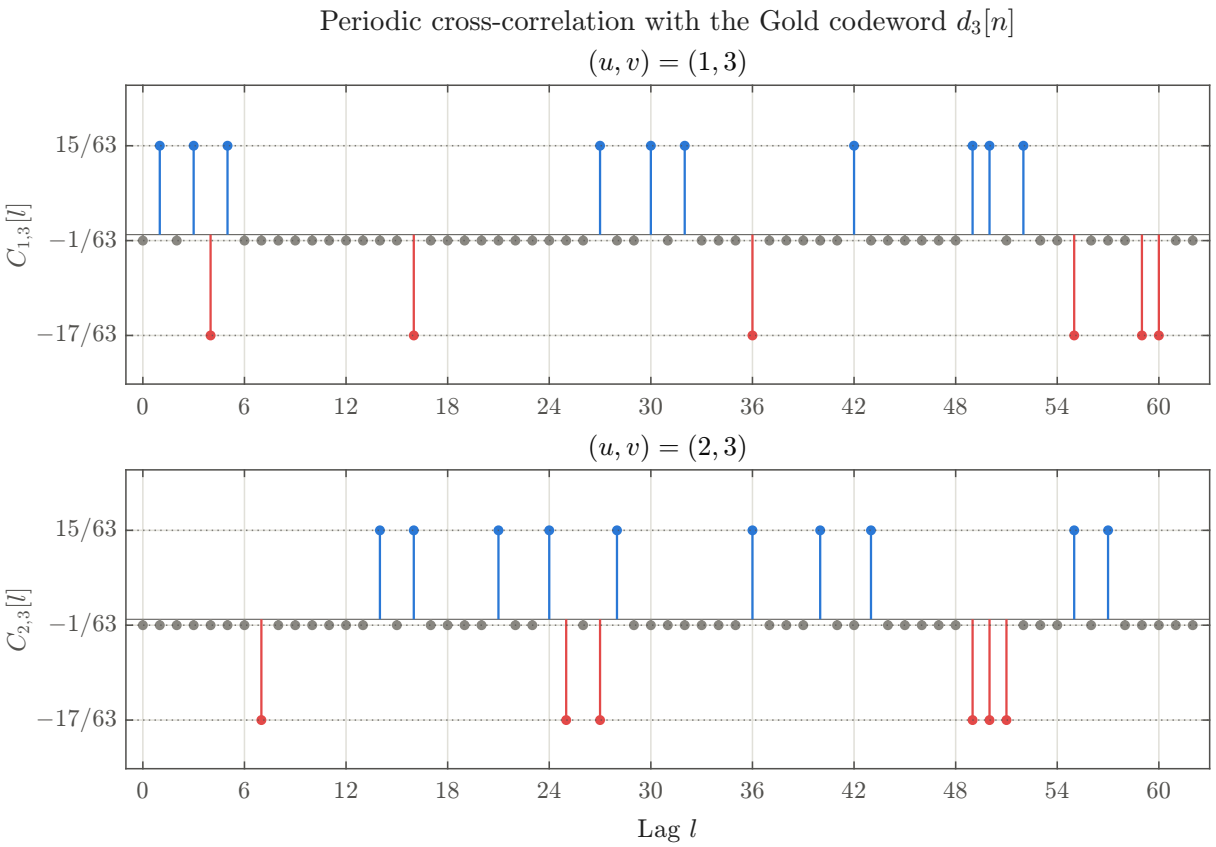


Figure 4.2: Periodic Cross-Correlation $C_{1,3}[l]$ (top) and $C_{2,3}[l]$ (bottom); blue: 15/63, gray: −1/63, red: −17/63

4.4 (d) Is the Answer in (c) Correct?

Please judge if your answer in (c) is correct or not? Please state the reason. (5%)

Yes, the answer in (c) is correct. I checked it in five ways.

1) Gold's theorem. $p(x)$ and $p'(x)$ are a preferred pair of degree $n = 6$. For a preferred pair, the periodic cross-correlation between the two m-sequences, or between any two codewords of the Gold family they generate, can only be

$$\frac{1}{N}\{-t(n),\ -1,\ t(n)-2\}, \qquad t(n) = 1 + 2^{\lfloor (n+2)/2 \rfloor} = 1 + 2^4 = 17, \tag{4.3}$$

i.e. $\{-17, -1, 15\}/63$, which is the set I got for $C_{1,3}$ and $C_{2,3}$.

2) How often each value appears. $C_{1,3}$ has to look like $C_{1,2}$. Using $d_3 = d_1 d_2$,

$$63\, C_{1,3}[l] = \sum_n \underbrace{d_1[n]\, d_1[n-l]}_{= d_1[n-k] \text{ for } l \neq 0}\, d_2[n-l], \tag{4.4}$$

so for $l \neq 0$ it equals the cross-correlation of d_1 and d_2 at some other lag, and at $l = 0$ it is $\sum_n d_2[n] = -1$. $C_{2,3}$ works the same way. For a preferred pair with $n = 6$ (decimation $q = 2^k + 1 = 5$, $e = \gcd(n, k) = 2$), the theory also says how many lags give each value:

$$-17 : 2^{n-e-1} - 2^{\frac{n-e}{2}-1} = 6, \qquad -1 : 2^n - 2^{n-e} - 1 = 47, \qquad 15 : 2^{n-e-1} + 2^{\frac{n-e}{2}-1} = 10. \tag{4.5}$$

$C_{1,2}$, $C_{1,3}$ and $C_{2,3}$ each give exactly 6, 47 and 10 (figure below).

3) The pair is preferred. Taking every fifth bit of s_1, i.e. $s_1[5n \bmod 63]$, gives s_2 shifted by 6 lags. So s_2 is the $q = 5$ decimation of s_1, and Gold's condition holds: $q = 2^2 + 1$ and $n/\gcd(n, 2) = 3$ is odd. I also tried $q = 13$, 21 and 23, and none of them gives s_2.

4) Sum check. For any two ±1 sequences, $\sum_l C_{u,v}[l] = \frac{1}{N}\big(\sum_n d_u[n]\big)\big(\sum_n d_v[n]\big)$. Each of the three sequences has 32 ones, so the sum should be $(-1)(-1)/63 = 1/63$. The counts in Table 4.4 give $(6 \times (-17) + 47 \times (-1) + 10 \times 15)/63 = 1/63$. ✓

5) The whole family. The Gold family has $N + 2 = 65$ codewords: d_1, d_2, and $d_1[n]\, d_2[n + k]$ for $k = 0, \ldots, 62$. I computed the cross-correlation for all $\binom{65}{2} = 2080$ pairs at all 63 lags, and every value was $-17/63$, $-1/63$ or $15/63$.

d_3 itself is not an m-sequence, so its own autocorrelation is not two-valued. Its 62 sidelobes are also -17, -1 and 15 (over 63), occurring 12, 38 and 12 times. A Gold family gives up the perfect $-1/63$ autocorrelation of a single m-sequence. In return it provides 65 codewords whose cross-correlation never exceeds $17/63$, and that is the property a CDMA system such as GPS relies on to tell users apart.

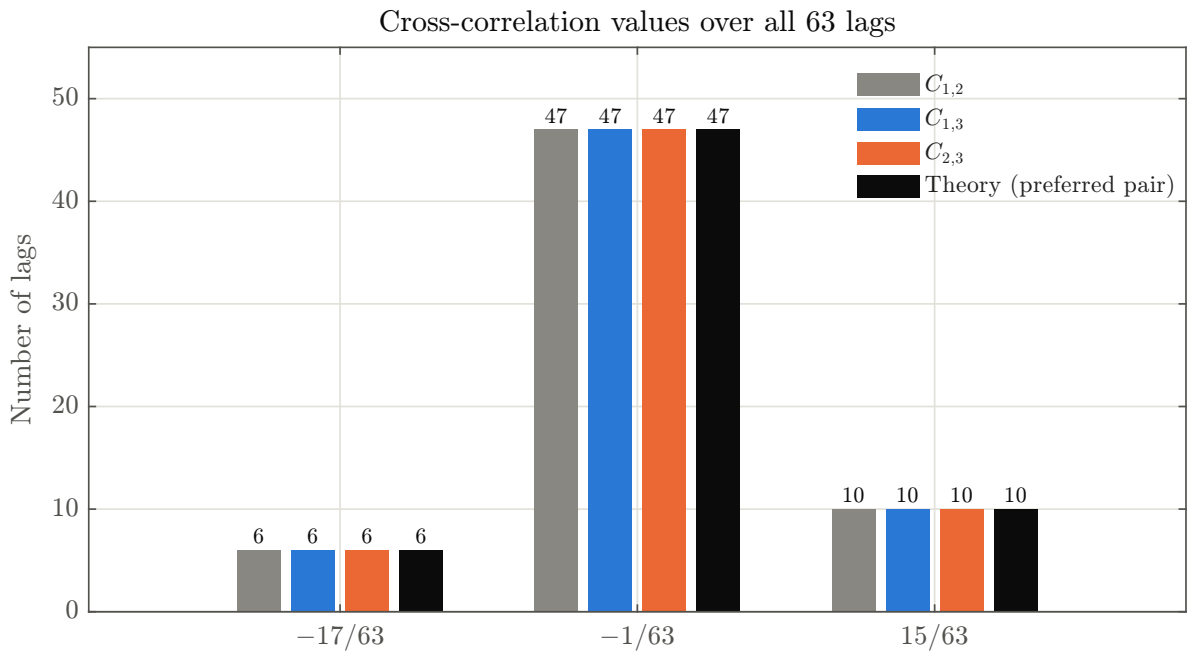


Figure 4.3: Number of Lags at Each Cross-Correlation Value Compared with the Preferred-Pair Theory

AI Tool Disclosure

Please disclose how you use AI tools and its version.

I used **Claude Opus 5.5** to help with this homework in the following four parts:

(1) Understanding the homework specification.

(2) Assisting with the MATLAB code.

(3) Plotting the figures.

(4) Writing this LaTeX report.